

Digital Forensics and genAI

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Large Language Models (LLMs)

- **advanced AI systems designed to understand and generate human-like output (text, images, ...)**
- **sophisticated neural networks that enable to process and produce coherent and contextually relevant language**
 - contain billions (or even trillions) of parameters
 - learned during training and represent the relationships and patterns within the data
 - allows these models to capture intricate details of language and generalize effectively across a wide range of contexts
 - trained on extensive datasets
 - transformer architectures and self-attention mechanisms
 - enabling the model to compute the relationships between words/tokens regardless of their distance

Using LLMs in investigative processes (I)

- **Enhance efficiency/effectiveness of various phases of an investigative process**
 - support in analyzing textual and visual data, automating tasks, and generating actionable insights
- **Log/data analysis for early detection**
 - parsing of large volumes of unstructured or semi-structured logs and interpretation of complex linguistic clues
 - suspicious activity flag
 - detection of potential security breaches, unauthorized access, IoC

Using LLMs in investigative processes (II)

- **LLMs can streamline the the collection phase identifying and organizing critically evidence, both textual and visual**
 - GPT-4 or newer excels in analyzing textual content, drafting reports, interpreting written evidence
 - special-purpose LLMs like **VisionLLM** combine natural language processing (NLP) with computer vision to analyze and categorize visual data such as photos and videos
 - can identify objects, faces, or text within images and classify them into relevant categories for further investigation

Using LLMs in investigative processes (III)

- **LLMs can automate and optimize tasks in the acquisition and preservation phases**
 - generate code for securely acquiring/preserving evidence
 - Support creation of automated workflows for extracting and storing data from devices or cloud services.
 - e.g. *AutoGen*, a framework that enables automation using AI agents, can work alongside LLMs to perform repetitive tasks such as data extraction, file categorization, or integrity checks
 - reduces the need for manual intervention, ensuring faster and more accurate acquisition processes

Using LLMs in investigative processes (IV)

- **LLMs can recover relevant data and conduct keyword-based searches with significant precision and efficiency**
 - retrieving data from diverse sources, including damaged or fragmented datasets
 - keyword-based searches with contextual understanding, uncovering relevant data even if the exact keyword isn't explicitly present
 - seamlessly connecting with forensic suites for automated analysis.
 - acting as an intelligent interface for querying databases and visualizing results

Using LLMs in investigative processes (V)

- **LLMs can support in deriving meaning and connecting disparate pieces of information**
 - process large datasets and establish connections between seemingly unrelated data points
 - identify trends, anomalies, or relationships that may not be immediately apparent
- support RAG (Retrieval-Augmented Generation), i.e. combination of LLMs with external knowledge sources to improve the accuracy and relevance of generated insights
 - e.g. investigators can query a corpus of evidence, and RAG systems will retrieve and analyze the most pertinent documents for review

Using LLMs in investigative processes (VI)

- **LLMs can aid creation of documentation of findings, often adhering to established standards**
 - draft comprehensive and well-structured final reports
 - create summaries
 - detailed evidence analyses
 - conclusions formatted for specific audiences
 - legal, managerial, technical...
 - according to international standard
 - e.g., ISO/IEC 27043:2015

Risks using LLMs (I)

■ Reliability

- bad actors actively deploy anti-forensic techniques (e.g., changing file extensions, embedding data using steganography, or using adversarial perturbations to confuse AI classifiers)
- LLMs must be tested against *adversarial robustness*.
- e.g. a defense attorney can challenge VisionLLM findings, if vulnerable to *adversarial patches* (for example, small pixel alterations invisible to humans but capable of causing an AI to misclassify a weapon as a harmless object)

Risks using LLMs (II)

■ Repetibility

- Forensics demands *determinism*.
- LLMs and agentic frameworks are probabilistic, running the exact same prompt twice through
- e.g. an *AutoGen* pipeline can yield two different execution traces or scripting behaviors
- Introduce deterministic sandboxing and verification engines
 - e.g *AutoDFBench* framework developed recently to test AI-generated forensic code against strict NIST standards

Risks using LLMs (III)

■ Recoverability

- Data carving performed by LLMs is "Semantic Carving"
- traditional data carving tools (like Scalpel or Foremost) look for rigid file headers and footers (e.g., \xFF\xD8\xFF for JPEGs)
- LLMs evaluate the linguistic and contextual flow between unallocated text fragments to predict how they should stitch back together, rather than physically fixing bad disk sectors
- even if well connected, the result may be *wrong!*

Risks using LLMs (IV)

■ Scalability

- Forensic data scales into terabytes of system logs, registry hives, and network captures
- RAG chunking mechanisms easily lose isolated (but potentially highly critical "needle-in-a-haystack" events
 - e.g., a single anomalous execution line hidden inside 50GB of standard syslog traffic.
- hybrid hierarchical RAG may be required
 - mixed regular rule-based security tools (SIEMs/YARA) to flag raw anomalies that creates structured knowledge graph, summarized/queried via an LLM

Risks using LLMs (V)

- **performance of LLMs heavily depends on the quality and specificity of their training data**
 - general-purpose LLMs may exhibit biases or inaccuracies when applied to forensic scenarios due to a lack of domain-specific training
 - to improve effectiveness, models should be fine-tuned with forensic-specific datasets,
 - digital logs, legal documents, cybercrime case studies
 - to ensure relevance and reduce bias
 - However, real forensic logs/files contain strictly confidential
 - authentic cybercrime cases may be legally blocked by compliance regulations (GDPR, HIPAA, corporate NDAs)
 - future forensic may relies on Synthetic Data Generation

Risks using LLMs (VI)

■ **Can't replicate Investigators' Intuition**

- investigators often rely on subtle contextual cues and years of expertise to make decisions that LLMs may not yet fully emulate
- effective use of LLMs requires a partnership between human expertise and machine intelligence, emphasizing the importance of human oversight

■ **Balance the benefits of automation and efficiency with the financial and resource demands of implementation**

- developing, training, and deploying LLMs for forensic applications can be resource-intensive requiring significant computational power and expertise
- fine-tuning models for specific domains further increases costs

Risks using LLMs (VII)

■ **Censorship and access to information**

- LLMs often require extensive datasets, which can lead to ethical concerns regarding data acquisition and usage
- investigative scenarios may involve private/classified information, raising data ownership/privacy concern
- strict ethical guidelines and governance structures are necessary to prevent misuse or overreach in data access
 - e.g. paste a compromised system's log file into a commercial service is an evidence invalidating data leak
 - trigger several legal and technical violations that a defense attorney will easily exploit to get that evidence not admissible in a court
 - *what processed this data? Did an employee view it? Was it cached on an unencrypted server? Was it intercepted?*

Risks using LLMs (VIII)

■ False or misleading Information

- "Black Box admissibility problem"
- generation of false, misleading, or inaccurate information
- determining accountability for errors becomes complex when AI-generated insights influence investigative decisions
- legal frameworks and ethical standards must address liability and ensure transparency in AI-assisted investigations
- only local, open-weights models (run on localized, completely air-gapped forensic workstations) are legally viable
 - LLM framed as "Investigative Lead Generator" rather than an expert witness